MIL-S-81551/2A(AS)

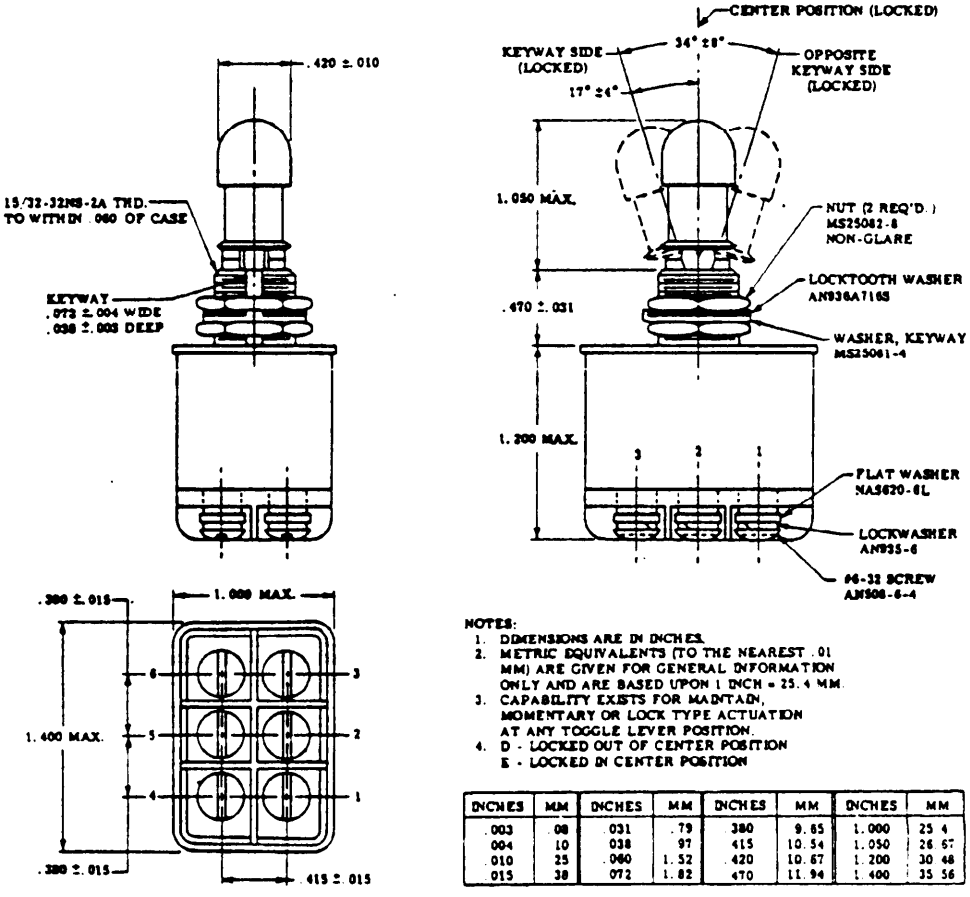
5 March 1970

* SUPERSEDING
MIL-S-81551/1(AS)
7 March 1968
(See below)

MILITARY SPECIFICATION SHEET

SWITCH, TOGGLE, DOUBLE POLE, LEVER LOCK, HERMETICALLY SEALED

The complete requirements for procuring the switches described herein shall consist of this document and the issue in effect of specification MIL-S-81551.



NOTES:
1. DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
3. CAPABILITY EXISTS FOR MAINTAIN, MOMENTARY OR LOCK TYPE ACTUATION AT ANY TOGGLE LEVER POSITION.
4. D - LOCKED OUT OF CENTER POSITION
   E - LOCKED IN CENTER POSITION

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.031	.79	.380	9.65	1.000	25 4
.004	.10	.038	.97	.415	10.54	1.050	26 67
.010	.25	.060	1.52	.420	10.67	1.200	30 48
.015	.38	.072	1.82	.470	11.94	1.400	35 56

TABLE I

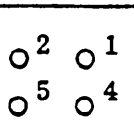
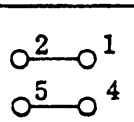
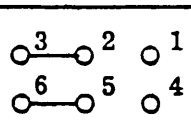
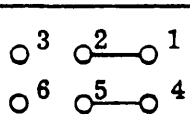
Circuit Diagram	Part Number	Circuit With Toggle Lever In			Available Locking Combination	Current Capacity Amperes 28 Volts D. C.			Current Capacity Amperes 115 Volts, 60 & 400 CY. A.C.		
		Opposite Keyway Side	Center	Keyway Side		Resistive	Inductive	Lamp Load	Resistive	Inductive	Lamp Load
I	M81551/2-482	None	Lock off	Mom. on	E	10	5	2	10	5	2
II	M81551/2-432	Lock on	None	Lock on	D	10	5	2	10	5	2

* (The 7 March 1968 issue was inadvertently numbered slash 1.

FSC 5930

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TABLE II

Circuit Diagram Number	Switch Function	Circuit With Toggle Lever In		
		Opposite Keyway Side	Center	Keyway Side
I	Two position 2PDT	No position		
II	Two position 2PDT		No position	

REQUIREMENTS:

Dimensions and configuration	See figure
Temperature characteristics	-65° F to 275° F non operating -65° F to 160° F operating
Shock type	MIL-E-5272, Procedure IV (50G)
Vibration grade	MIL-E-5272, Procedure XII Curve A (10G)
Weight	0.30 pound maximum
Material	All external metal parts shall be corrosion resistant

OPERATING CHARACTERISTICS:

1. Insulation resistance	MIL-STD-202, Method 302 1,000 megohms, 500 VDC
2. Dielectric withstanding voltage	MIL-STD-202, Method 301
A) Atmosphere pressure	1,000 VRMS, max. leakage .5 MA
B) At 80,000 feet	400 VRMS, max. leakage .5 MA
3. Seal test	1×10^{-8} CC HE/SEC per atmosphere maximum leak rate
4. Mechanical endurance	50,000 cycles
5. Electrical endurance	25,000 cycles at all current ratings
6. Electrical ratings	See Table I
7. Short circuit capability	Capable of close in on 600 amperes calibrated circuit 5 consecutive times
8. Force to release locking mechanism	2 to 5 pounds

Custodian:
Navy - AS

Preparing activity:
Navy - AS
Project No. 5930-N279-2

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